

Markov Processes: Homework 2

Due Wed Sept 9 by 11:59pm on Gradescope

Format. Plots, code, and write-up follow the Homework 1 conventions, and code goes in the same “Markov2026” repository labeled by problem. New this week, normalize histograms as empirical PDFs, state the bin width, and overlay the theory. For acceptance–rejection, also report the empirical acceptance fraction beside $1/c$ and the mean time per accepted sample.

Optional problems. Problems after the END box, and parts marked optional, are not graded.

1. Discrete inverse transform and the cost of a search.

- (a) Let $Z \in \{1, 2, 3\}$ with $P(Z = 1) = 0.05$, $P(Z = 2) = 0.15$, $P(Z = 3) = 0.80$. Give the cumulative thresholds and a three line pseudo-code that returns Z from a single $U \sim U(0, 1)$.
- (b) Count the inequalities checked by your pseudo-code (an “else” is not a comparison). Compute the expected number of comparisons for the order above and for the order that minimizes it. State the rule in one sentence.
- (c) Let X be geometric on $\{1, 2, 3, \dots\}$ with $P(X = k) = p(1-p)^{k-1}$, so that $F(k) = 1 - (1-p)^k$. Show that solving $F(k) \geq U$ for the smallest such k gives

$$X = \left\lceil \frac{\ln(1-U)}{\ln(1-p)} \right\rceil.$$

2. Inverse transform on the whole line, and a law of large numbers that fails. Let X

have the standard Cauchy density $f(x) = \frac{1}{\pi(1+x^2)}$, $x \in \mathbb{R}$.

- (a) Derive F and F^{-1} , and give two line pseudo-code for sampling X .
 - (b) Show that the median of $|X|$ is 1 while $E|X| = \infty$. Explain in one or two sentences how a typical draw can be of order one while the mean fails to exist.
- ### 3. Acceptance–rejection with an exponential envelope. Let $X \sim \text{Gamma}(2, 1)$ with density $f(x) = xe^{-x}$ for $x \geq 0$. The CDF is $F(x) = 1 - (1+x)e^{-x}$, which cannot be inverted in closed form, so we use rejection with the proposal $g(x) = \lambda e^{-\lambda x}$.
- (a) Take $\lambda = 1/2$. Show that the smallest valid envelope constant is $c = 4/e$, state the theoretical acceptance rate, and show that the acceptance test simplifies to $U < \frac{e}{2}Xe^{-X/2}$.
 - (b) Give succinct pseudo-code (single while loop), sampling the proposal by inversion.
 - (c) Implement this for $N = 10^4$ accepted samples. Plot the normalized histogram with $f(x)$ overlaid, report the empirical acceptance fraction against $1/c$, and report the mean time per accepted sample. Then repeat with $\lambda = 0.2$, for which $c = 1/(0.16e) = 2.2993$, and comment on the difference.
- ### 4. Ion channel dwell times and mixture sampling. A patch clamp record of a single channel is a sequence of closed and open intervals. Markov gating models predict that dwell times in a set of states are mixtures of exponentials, with one component per state in the set (Colquhoun and Hawkes, *Phil. Trans. R. Soc. Lond. B* 1982). Take a channel with a brief and a long closed state, so that the closed dwell time T has density

$$f(t) = a\lambda_f e^{-\lambda_f t} + (1-a)\lambda_s e^{-\lambda_s t}, \quad t \geq 0,$$

with $a = 0.9$, $\lambda_f = 1000 \text{ s}^{-1}$ and $\lambda_s = 10 \text{ s}^{-1}$.

- (a) Verify that f is a density and compute $E[T]$, $\text{Var}(T)$, and the coefficient of variation $\text{CV} = \sqrt{\text{Var}(T)}/E[T]$. Compare the CV with that of a single exponential and say in one or two sentences what the comparison tells you.
- (b) Give a three step composition algorithm using one Bernoulli draw and one inverse transform draw, and show by conditioning on the component that it returns a sample from f .

- (c) Generate 10^5 dwell times by composition. Plot the normalized histogram on semilog vertical axes with $f(t)$ overlaid, so that both components are visible. Report the empirical mean and the empirical $P(T > 50 \text{ ms})$ against theory.

5. **Simulation Assignment: growth by redirection, and an average nobody obeys.** Hartle, Krapivsky, Redner and Zhang (*arXiv:2604.01540*, 2026) study trees grown by isotropic redirection, in which each new node picks an existing node uniformly at random and then attaches to a randomly chosen neighbor of that node. Almost every node ends up a leaf, and the number C of core nodes (degree at least two) grows sublinearly, $C \sim N^\mu$ with $\mu \approx 0.566$. Their appendix strips the model down to a two population caricature that tracks only the core count,

$$C(N+1) = \begin{cases} C(N) + 1, & \text{with probability } \mu C(N)/N, \\ C(N), & \text{otherwise,} \end{cases} \quad C(3) = 1,$$

with $0 < \mu \leq 1$. Each step is one Bernoulli draw, so this is a sampling problem. Take $\mu = 1$ throughout, the marginal case, for which one can show that $z = C(N)/N$ has the limiting density

$$h(z) = 2(1 - z), \quad 0 < z < 1.$$

- (a) Check that h is a density and compute $E[z]$, $\text{Var}(z)$, and the ratio of standard deviation to mean.
- (b) Sample h by inversion. Derive H and H^{-1} , give two line pseudo-code, and generate 10^5 samples. Check the sample mean against part (a).
- (c) Simulate the growth process at $\mu = 1$ out to $N = 10^4$ for $R = 10^3$ independent realizations, vectorizing across realizations. By this I mean that the update at step n must draw a single length R vector of Bernoulli variates: a loop over realizations wrapped around a loop over steps is 10^7 interpreted iterations and will not finish in a sitting. Plot the normalized histogram of $z = C/N$ with $h(z)$ overlaid, and report the empirical mean and the empirical ratio of standard deviation to mean against your answers in part (a). Report the smallest and largest core you saw, and comment in two or three sentences on what the average core size does and does not tell you about a single network.

END

6. **Optional Practice: implementing the discrete samplers.** Problem 1 derived the categorical and geometric inversion samplers; here you run them.

- (a) Implement both samplers. For the categorical variable generate $M = 10^5$ samples and plot the empirical frequencies against the theoretical probabilities. For the geometric with $p = 0.01$ generate $M = 10^5$ samples by the closed form, plot the empirical PMF on semilog axes against $p(1 - p)^{k-1}$, and check the empirical mean and standard deviation against $1/p$ and $\sqrt{1 - p}/p$.

7. **Optional Practice: a law of large numbers that fails, on screen.** The standard Cauchy of Problem 2 is sampled by $X = \tan(\pi(U - 1/2))$.

- (a) Generate 10^6 Cauchy samples by inversion and plot the running sample mean against N on a logarithmic horizontal axis, together with the running mean of 10^6 Exponential(1) samples on the same axes. Rerun with a different seed and describe what changes.

8. **Optional Practice: why acceptance–rejection works, and what it costs.**

- (a) With $Y \sim g$ and $U \sim U(0, 1)$ independent and $c \geq \sup f/g$, let A be the event $U < f(Y)/(cg(Y))$. Show that $P(A) = 1/c$ and $P(Y \leq x | A) = F(x)$.

- (b) Deduce the distribution of the number of proposals per accepted sample, its mean, and the average number of uniforms consumed, evaluating each for Problem 3 at $\lambda = 1/2$.
 - (c) For the Gamma target of Problem 3, show that the optimal envelope constant is $c(\lambda) = 1/[e\lambda(1 - \lambda)]$, minimized at $\lambda = 1/2$, and that no finite c exists once $\lambda \geq 1$.
 - (d) For the dwell time density of Problem 4 with proposal $g(t) = \lambda_s e^{-\lambda_s t}$, find the smallest valid c and the acceptance rate, then show that $g(t) = \lambda_f e^{-\lambda_f t}$ admits no finite c .
9. **Optional Practice: the standard normal by rejection.** Let $|Z|$ have the half normal density $f(x) = \sqrt{2/\pi} e^{-x^2/2}$ for $x \geq 0$, and take the proposal $g(x) = e^{-x}$.
- (a) Find the optimal c and the acceptance rate, and show that the acceptance test reduces to $U < e^{-(X-1)^2/2}$.
 - (b) Give pseudo-code for a standard normal variate, explaining the step from $|Z|$ to Z .
 - (c) Generate 10^5 samples, plot the normalized histogram against the standard normal density, report the acceptance fraction and time per accepted sample, and compare with the built in generator.
10. **Optional Practice: heavy tails, two ways.**
- (a) The Cauchy of Problem 2 has no MGF, but its characteristic function is $\varphi_X(t) = E[e^{itX}] = e^{-|t|}$. Taking that for granted, show that the sample mean of n i.i.d. standard Cauchy variables is again standard Cauchy, and say what that means for Problem 7.
 - (b) The core degree distribution in Problem 5 decays as $p_k \propto k^{-(1+\mu)}$, $k \geq 2$. Show that $E[k] = \infty$ for $\mu < 1$, then use $\sum_k k N_k = 2(N - 1)$ with $N_k \sim N^\mu k^{-(1+\mu)}$ to explain why the average degree stays near two.
 - (c) Plot the running maximum $M_N = \max_{i \leq N} X_i$ of the Problem 7 Cauchy sample against N on log-log axes, predicting the slope first.
 - (d) Sample k by flooring a continuous Pareto variate, $k = \lfloor 2(1 - U)^{-1/\mu} \rfloor$ with $\mu = 0.55$, and plot $E[k]$ from 10^6 samples against a truncation $k_{\max} = 10^3, 10^4, 10^5, 10^6$.
11. **Optional Practice: the redirection core for general μ .** Problem 5 ran the two population process at $\mu = 1$. The regime the paper cares about is $\mu < 1$, where the core is a vanishing fraction of the tree.
- (a) Show that $E[C(N)] = 2\Gamma(N + \mu)/[\Gamma(3 + \mu)\Gamma(N)] \sim AN^\mu$ with $A = 2/\Gamma(3 + \mu)$, and check the case $\mu = 1$ against Problem 5(a).
 - (b) Repeat the Problem 5 simulation at $\mu = 0.55$, a value the paper derives for one of its tractable variants, for $N = 10^2, 10^3, 10^4, 10^5$. Plot $E[C(N)]$ against N on log-log axes with the part (a) prediction, and report the fitted exponent and the ratio of standard deviation to mean at each N .
12. **Optional Practice: a gating trajectory.** (Preview of continuous time chains.) A channel alternates between open and closed with independent Exponential(α) open durations and Exponential(β) closed durations, starting closed at $t = 0$.
- (a) Simulate a trajectory out to $T = 100$ s with $\alpha = 200$ s $^{-1}$ and $\beta = 20$ s $^{-1}$ by drawing alternating dwell times with inversion, and plot the first 0.5 s as a square wave.
 - (b) Compare the estimated fraction of time open with $\beta/(\alpha + \beta)$, and the estimated number of openings per second with $\alpha\beta/(\alpha + \beta)$.
 - (c) Histogram the number of openings in windows of length 0.1 s, overlay a Poisson PMF with the matching mean, and comment on where the fit fails and why.